

Diagnostic report: Mg I

Computed levels: 25

Matched to NIST: 15

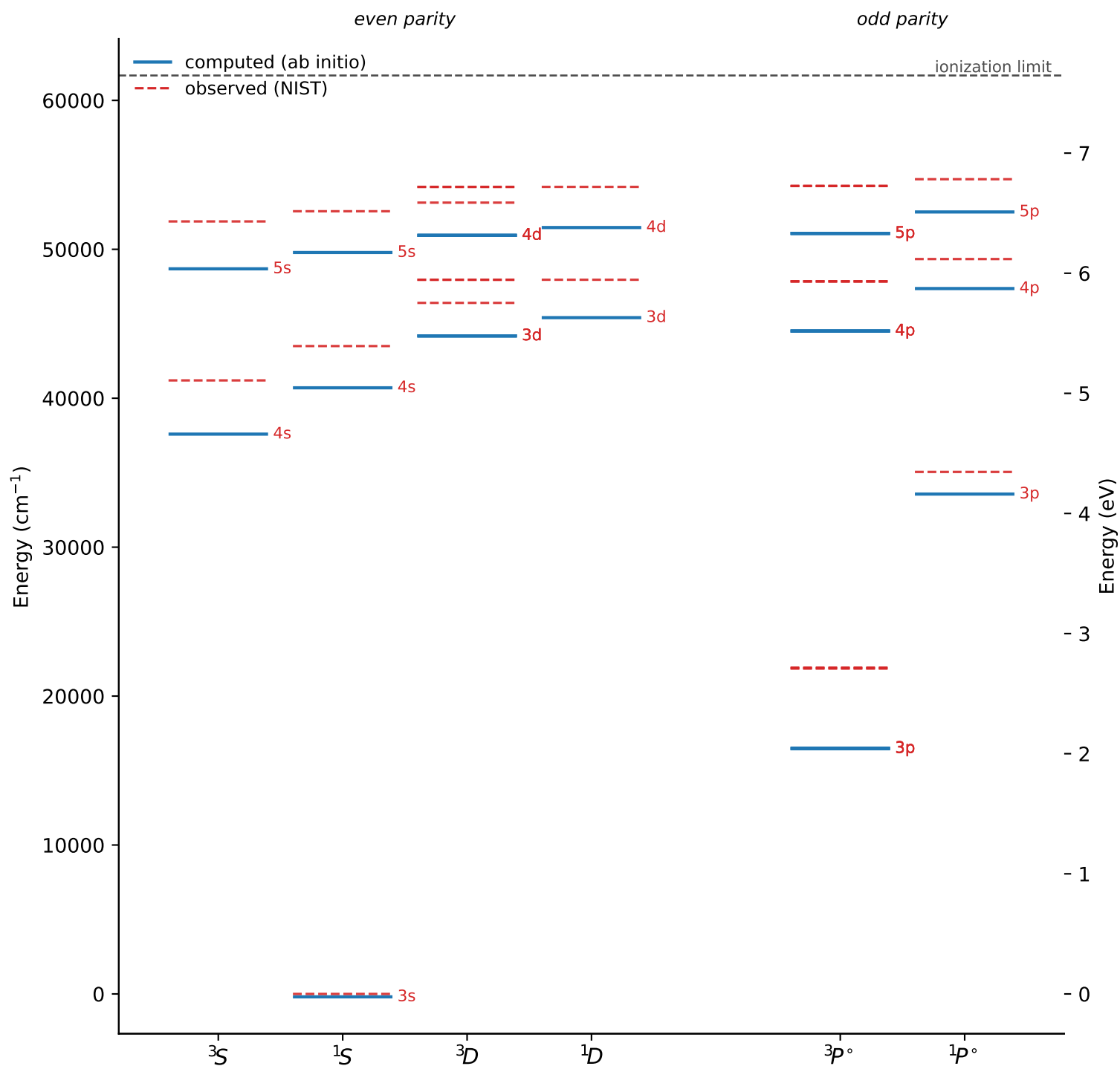
Computed E1 lines: 52

Pre-fit RMS ($E_{\text{calc}} - E_{\text{obs}}$): 24186 cm^{-1} (ab initio, no RCE fit yet)

Strongest computed lines:

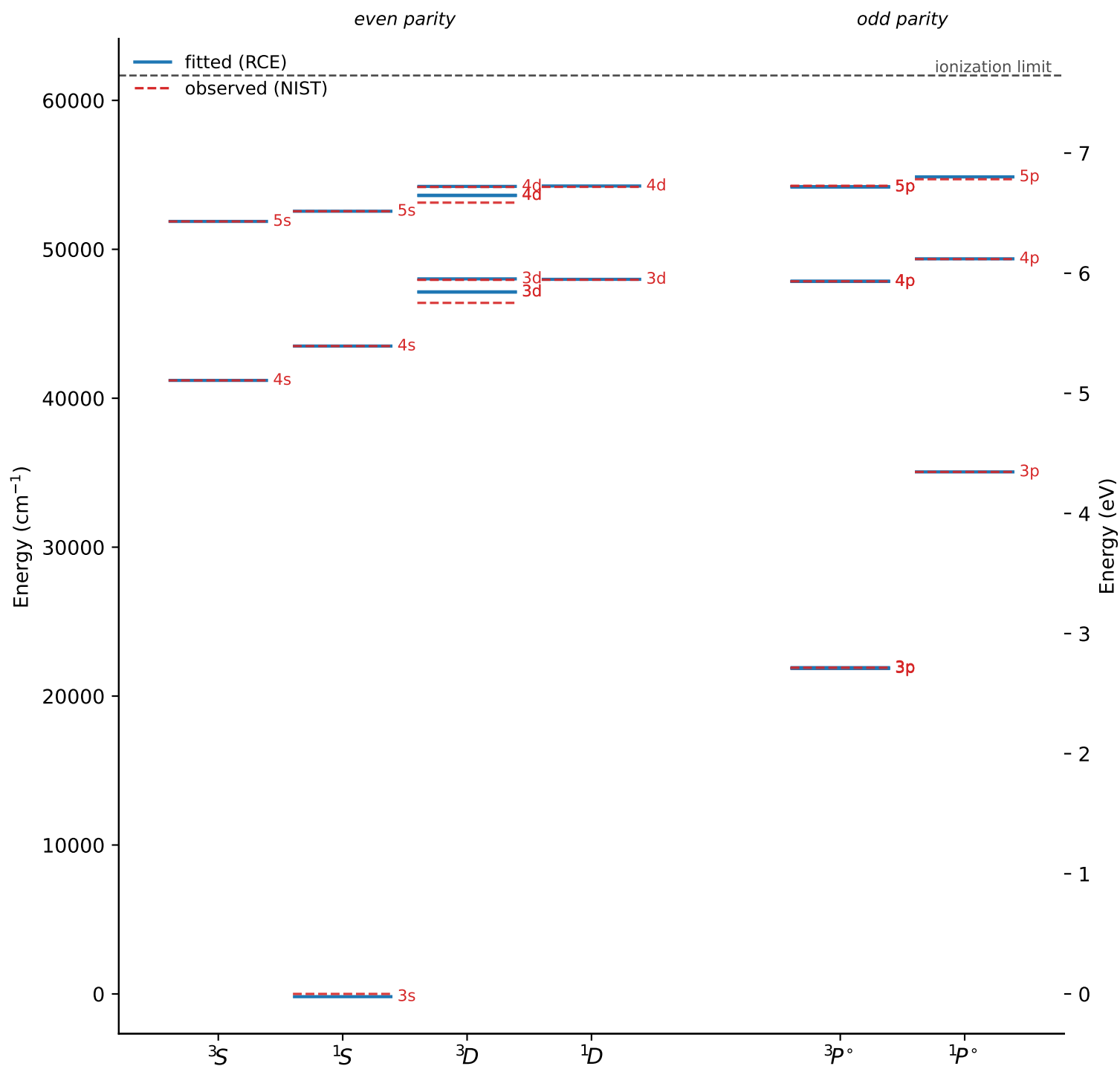
lambda(A)	log gf	upper term
24404.97	+0.595	(2S) 1P
42031.78	+0.507	(2S) 3P
3612.66	+0.470	(2S) 3P
15581.33	+0.424	(2S) 3P
8441.88	+0.421	(2S) 1P
2962.04	+0.393	(2S) 1P
14430.30	+0.382	(2S) 3P
42069.26	+0.285	(2S) 3P

Mg I: term diagram - ab initio vs observed



One column per term; levels labelled by running orbital (red). Solid = computed (ab initio), dashed = observed (NIST).

Mg I: term diagram - fitted vs observed



One column per term; levels labelled by running orbital (red). Solid = fitted (RCE), dashed = observed (NIST).

Mg I: levels — ab initio & fitted vs observed (cm⁻¹)

config	term	J	E_obs	E_calc	Δ_{abinit}	E_fit	Δ_{fit}
3s2	¹ S	0	0.0	-195.0	-195.0	-183.1	-183.1
3s.3p	³ P°	0	21850.4	16463.0	-5387.4	21850.3	-0.1
3s.3p	³ P°	1	21870.5	16477.0	-5393.5	21870.5	+0.0
3s.3p	³ P°	2	21911.2	16505.0	-5406.2	21911.1	-0.1
3s.3p	¹ P°	1	35051.3	33566.0	-1485.3	35053.2	+1.9
3s.4s	³ S	1	41197.4	37590.0	-3607.4	41197.4	+0.0
3s.4s	¹ S	0	43503.3	40700.0	-2803.3	43503.9	+0.6
3s.3d	³ D	2	46403.1	44185.0	-2218.1	47126.3	+723.2
3s.3d	³ D	1	47957.1	44185.0	-3772.1	47143.4	-813.7
3s.3d	³ D	3	47957.0	44185.0	-3772.0	48015.9	+58.9
3s.4p	³ P°	0	47841.1	44511.0	-3330.1	47841.0	-0.1
3s.4p	³ P°	1	47844.4	44514.0	-3330.4	47844.3	-0.1
3s.4p	³ P°	2	47851.2	44519.0	-3332.2	47851.1	-0.1
3s.3d	¹ D	2	47957.0	45411.0	-2546.0	47985.3	+28.3
3s.4p	¹ P°	1	49346.7	47365.0	-1981.7	49361.9	+15.2
3s.5s	³ S	1	51872.5	48690.0	-3182.5	51872.5	+0.0
3s.5s	¹ S	0	52556.2	49785.0	-2771.2	52556.3	+0.1
3s.4d	³ D	1	54192.3	50937.0	-3255.3	53608.9	-583.4
3s.4d	³ D	2	53134.6	50937.0	-2197.6	53629.8	+495.2
3s.4d	³ D	3	54192.3	50937.0	-3255.3	54224.6	+32.3
3s.5p	³ P°	0	54248.8	51066.0	-3182.8	54191.3	-57.5
3s.5p	³ P°	1	54250.1	51067.0	-3183.1	54192.6	-57.5
3s.5p	³ P°	2	54252.7	51070.0	-3182.7	54195.2	-57.5
3s.4d	¹ D	2	54192.3	51462.0	-2730.3	54251.4	+59.1
3s.5p	¹ P°	1	54706.5	52508.0	-2198.5	54862.4	+155.9

Mg I: level residuals (ab initio RMS = 3311, fit RMS = 272 cm⁻¹)

